

Backend Update Patch — Subscription, Quota & Enterprise B2B

Tanggal: 2026-06-13 **Scope:** Subscription Plans, Payments (Midtrans), Tenant Quota, Manual Override, System Audit Trail **Status:** Belum di-commit (working tree)

Dokumen ini menjelaskan seluruh perubahan backend yang belum di-commit beserta dokumentasi API detail untuk tim Frontend.

1. Ringkasan Perubahan

Area	Perubahan
Enum Tier	Menambahkan tier baru <code>BASIC</code> (urutan: <code>FREE</code> , <code>BASIC</code> , <code>PRO</code> , <code>CUSTOM</code>)
Workspace Quota	Tenant kini punya limit jumlah workspace (<code>maxWorkspaces</code>), divalidasi saat pembuatan workspace
Price Anchoring & Promo	<code>SubscriptionPlan</code> mendukung harga coret (<code>originalPrice</code>) + campaign promo (<code>campaignName</code> , <code>campaignEndDate</code>)
B2B Quote-to-Cash	Tier <code>CUSTOM</code> tidak lagi self-serve. Invoice <code>CUSTOM</code> diterbitkan Super Admin
Manual Override	Super Admin bisa mem-paksa limit tenant ke <code>CUSTOM</code> tanpa melalui pembayaran
System Audit Trail	Tabel baru <code>systemAuditLog</code> untuk mencatat aksi level admin/sistem
Grand Reset	Endpoint internal untuk reset semua tenant ke <code>FREE</code>

2. Perubahan Database (Prisma Schema)

2.1 Enum `TenantTier`

```
enum TenantTier {  
  FREE  
  BASIC    // ← BARU  
  ...  
}
```

```
PRO
CUSTOM
}
```

2.2 Model SubscriptionPlan (field baru)

Field	Tipe	Keterangan
maxWorkspaces	Int @default(0)	Batas jumlah workspace. 0 = unlimited
originalPrice	Decimal? (12,2)	Harga asli sebelum diskon (price anchoring). Nullable
campaignName	String?	Nama event promo, mis. "Early Bird Launch". Nullable
campaignEndDate	DateTime?	Batas waktu promo (ISO). Nullable

Catatan: `durationDays` kini menerima nilai 0 (penanda durasi unlimited/selamanya).

2.3 Model Tenant (field baru)

Field	Tipe	Keterangan
maxWorkspaces	Int @default(3)	Limit workspace aktif tenant. 0 = unlimited

2.4 Model baru SystemAuditLog (terpisah dari Tenant)

```
model SystemAuditLog {
  id          String  @id @default(uuid()) @db.Uuid
  adminId     String  @map("admin_id") @db.Uuid
  action      String
  targetId    String? @map("target_id") // mis. tenantId yang diubah
  payload     Json?   // data perubahan
  createdAt   DateTime @default(now()) @map("created_at")

  @@index([adminId])
  @@index([action])
  @@map("system_audit_logs")
}
```

2.5 Migrasi

prisma/migrations/20260612160807_add_max_workspaces/

```
prisma/migrations/20260612164605_add_promo_fields/  
prisma/migrations/20260612171926_add_system_audit_log/  
prisma/migrations/20260612174233_add_basic_tier_enum/
```

Jalankan `npx prisma migrate deploy (prod)` atau `npx prisma migrate dev / npx prisma db push (dev)` agar DB sinkron.

3. ⚠ Breaking Changes (Penting untuk FE)

1. `POST /api/payments/checkout` tidak lagi menerima `requestedStorageGb`. Field tersebut dihapus dari body. Jika FE masih mengirimnya, `ValidationPipe` (whitelist) akan menolak request dengan `400`.
2. **Tier `CUSTOM` diblokir di checkout self-serve.** `POST /api/payments/checkout` dengan `tierToUpgrade: "CUSTOM"` mengembalikan `400`. `CUSTOM` kini hanya lewat invoice admin.
3. **Pembuatan workspace bisa gagal karena kuota.** `POST /api/tenants/:tenantId/workspaces` mengembalikan `400` bila limit `maxWorkspaces` tercapai.

4. Format Response Global

Semua response sukses dibungkus oleh `ResponseInterceptor` :

```
{  
  "statusCode": 200,  
  "status": "success",  
  "data": { /* payload aktual di sini */ }  
}
```

Response error mengikuti format standar NestJS `HttpException` (`statusCode` , `message` , `error`). **Base URL semua endpoint diawali prefik `/api`.**

5. Dokumentasi API

5.1 Subscription Plans — Admin CRUD

Auth: `Bearer JWT` + **Super Admin**. Field baru pada body Create/Update: `maxWorkspaces` , `originalPrice` , `campaignName` , `campaignEndDate` .

POST /api/admin/plans — Buat plan

Body (CreateSubscriptionPlanDto):

Field	Tipe	Wajib	Aturan
tier	enum FREE BASIC PRO CUSTOM	✓	
name	string	✓	
description	string	—	
basePrice	number	✓	>= 0
maxUsers	int	✓	>= 0
baseStorage	number (bytes)	✓	>= 0
maxWorkspaces	int	✓	>= 0 (0 = unlimited)
pricePerGb	number	—	>= 0
isActive	boolean	—	default true
originalPrice	number	—	>= 0 — harga coret
campaignName	string	—	nama promo
campaignEndDate	string (ISO 8601)	—	batas promo
durationDays	int	—	>= 0 (0 = selamanya), default 30

Contoh request:

```
{
  "tier": "BASIC",
  "name": "Basic Plan",
  "basePrice": 99000,
  "originalPrice": 149000,
  "campaignName": "Early Bird Launch",
  "campaignEndDate": "2026-12-31T23:59:59.000Z",
  "maxUsers": 10,
  "maxWorkspaces": 5,
  "baseStorage": 5368709120,
  "durationDays": 30
}
```

Contoh data pada response (Decimal/BigInt dikonversi ke number; originalPrice

```
number|null );
```

```
{
  "id": "uuid",
  "tier": "BASIC",
  "name": "Basic Plan",
  "basePrice": 99000,
  "originalPrice": 149000,
  "campaignName": "Early Bird Launch",
  "campaignEndDate": "2026-12-31T23:59:59.000Z",
  "maxUsers": 10,
  "maxWorkspaces": 5,
  "baseStorage": 5368709120,
  "pricePerGb": 0,
  "durationDays": 30,
  "isActive": true
}
```

PATCH /api/admin/plans/:id — Update plan

Body (`UpdateSubscriptionPlanDto`): semua field di atas, **semuanya opsional** (hanya field yang dikirim yang di-update).

Catatan FE — Price Anchoring: Tampilkan harga coret bila `originalPrice != null && originalPrice > basePrice`. Tampilkan badge promo bila `campaignName != null` dan (opsional) `campaignEndDate` belum lewat.

5.2 Checkout Standard (Self-Service) — BASIC & PRO

POST /api/payments/checkout

Auth: `Bearer JWT` (HybridAuthGuard). Body (`CreateCheckoutDto`):

Field	Tipe	Wajib	Aturan
<code>tenantId</code>	string (uuid v4)	✓	
<code>tierToUpgrade</code>	enum <code>BASIC PRO</code>	✓	<code>FREE</code> & <code>CUSTOM</code> ditolak

`requestedStorageGb` **DIHAPUS** — jangan dikirim lagi.

Response `data` :

```
{ "token": "midtrans-snap-token", "redirect_url": "https://app.midtrans.com/snap/.
```


Error:

Kondisi	Status	Pesan
tierToUpgrade: "FREE"	400	Cannot checkout for FREE tier
tierToUpgrade: "CUSTOM"	400	Custom plan requires sales negotiation. Please contact support.
Tenant sudah di tier yang sama	400	Tenant is already on the {tier} plan
Ada transaksi PENDING	400	pesan terkait pending transaction

BASIC diperlakukan identik dengan **PRO** (paket self-service standar).

5.3 Custom Invoice (B2B Quote-to-Cash) — BARU

POST /api/admin/invoices/custom

Auth: Bearer JWT + **Super Admin**. Diterbitkan admin untuk deal Enterprise yang sudah dinegosiasi. Body (CreateCustomInvoiceDto) — semua wajib:

Field	Tipe	Aturan
tenantId	string (uuid v4)	
amount	number	>= 0 (nominal hasil negosiasi, IDR)
maxUsers	int	>= 0
maxWorkspaces	int	>= 0
maxStorageGb	int	>= 0 (dalam GB)
durationDays	int	>= 1
description	string	catatan invoice

Response data :

```
{
  "id": "uuid",
  "orderId": "CUSTOM-xxxxxxx-1700000000000",
  "tenantId": "uuid",
  "grossAmount": 5000000,
  "status": "PENDING",
  "tierToUpgrade": "CUSTOM",
  "snapshotPlan": {
```

```

    "name": "Enterprise Custom Plan",
    "basePrice": 50000000,
    "maxUsers": 100,
    "maxWorkspaces": 50,
    "maxStorage": 536870912000,
    "durationDays": 365,
    "description": "Enterprise annual contract"
  },
  "token": "midtrans-snap-token",
  "paymentUrl": "https://app.midtrans.com/snap/..."
}

```

Error: `404` bila tenant tidak ditemukan.

Alur FE Admin: Kirim spesifikasi → terima `paymentUrl` → berikan/kirim link ke customer untuk membayar. Saat customer membayar, webhook Midtrans otomatis meng-upgrade tenant ke CUSTOM dengan limit dari `snapshotPlan`.

5.4 Manual Override Limit Tenant — BARU

PATCH `/api/admin/tenants/:id/custom-limits`

Auth: `Bearer JWT` + **Super Admin**. Memaksa tenant ke tier `CUSTOM` **tanpa** pembayaran (akun komplimen/negosiasi). Tercatat di `SystemAuditLog`. Path param: `id` = tenantId (uuid v4). Body (`OverrideTenantLimitsDto`) — semua **opsional** (hanya yang dikirim yang diubah):

Field	Type	Aturan
<code>maxUsers</code>	int	<code>>= 0</code>
<code>maxWorkspaces</code>	int	<code>>= 0</code>
<code>maxStorageGb</code>	int	<code>>= 0</code> (GB; disimpan sebagai bytes)
<code>validUntil</code>	string (ISO 8601)	tanggal akhir tier

Contoh request:

```
{ "maxUsers": 200, "maxWorkspaces": 100, "maxStorageGb": 1024, "validUntil": "2027
```

Response `data` : objek tenant (tier dipaksa `CUSTOM`, `usedStorage` & `maxStorage` dikonversi ke number). Error `404` bila tenant tidak ditemukan.

5.5 Workspace Quota Enforcement

POST /api/tenants/:tenantId/workspaces

Auth: Bearer JWT + permission workspace.create . Perilaku baru: sebelum membuat workspace, sistem mengecek checkWorkspaceLimit . Error baru:

Kondisi	Status	Pesan
Jumlah workspace aktif >= maxWorkspaces	400	Workspace limit reached. Your {tier} plan allows up to {n} workspaces. Current: {m}.

Bila `maxWorkspaces = 0` , dianggap unlimited (tidak dibatasi). **Saran FE:** Tampilkan sisa kuota workspace & sembunyikan/disable tombol "Create Workspace" bila kuota habis. Tangani error `400` di atas dengan pesan upgrade plan.

6. Perubahan Logika Internal (tanpa kontrak API baru)

- **Webhook Midtrans (`handleWebhook`)** — saat settlement, limit tenant (`maxUsers` , `maxWorkspaces` , `maxStorage`) kini dibaca seragam dari `snapshotPlan` untuk semua tier (logika `requestedStorageGb` untuk CUSTOM dihapus). Idempotency (skip transaksi yang sudah final) & validasi signature SHA512 tetap dipertahankan.
- **Snapshot saat checkout** — `snapshotPlan` kini juga merekam `maxStorage` (unified), `maxWorkspaces` , `originalPrice` , dan `campaignName` untuk keperluan invoice/riwayat.
- **Atomicity** — `overrideCustomLimits` dan `resetAllTenantsToFreePlan` membungkus update + pencatatan audit log dalam satu `prisma.$transaction` (all-or-nothing).

7. Daftar File yang Berubah

Modif ed:

```
prisma/schema.prisma
src/common/services/tenant-limits.service.ts      (+ checkWorkspaceLimit)
src/workspaces/workspaces.service.ts               (panggil checkWorkspaceLimit)
src/subscription-plans/dto/create-subscription-plan.dto.ts
src/subscription-plans/dto/update-subscription-plan.dto.ts
src/subscription-plans/subscription-plans.service.ts
src/payments/dto/create-checkout.dto.ts            (hapus requestedStorageGb)
src/payments/payments.controller.ts
src/payments/payments.module.ts                    (daftarkan AdminPaymentsControll
src/payments/payments.service.ts                   (block CUSTOM, custom invoice, w
src/tenants/tenants.module.ts                       (daftarkan AdminTenantsControll
```


src/tenants/tenants.service.ts	(overrideCustomLimits + audit)
src/internal/internal.controller.ts	(route grand reset)
src/internal/internal.service.ts	(resetAllTenantsToFreePlan)

New (untracked):

src/payments/admin-payments.controller.ts	(POST /api/admin/invoices/custom)
src/payments/dto/create-custom-invoice.dto.ts	
src/tenants/admin-tenants.controller.ts	(PATCH /api/admin/tenants/:id/limits)
src/tenants/dto/override-tenant-limits.dto.ts	
prisma/migrations/20260612160807_add_max_workspaces/	
prisma/migrations/20260612164605_add_promo_fields/	
prisma/migrations/20260612171926_add_system_audit_log/	
prisma/migrations/20260612174233_add_basic_tier_enum/	

8. Checklist Deploy

- ☐ Jalankan migrasi DB (`prisma migrate deploy`).
- ☐ Pastikan plan `FREE` ada di tabel `subscription_plans` (dibutuhkan Grand Reset & default).
- ☐ **Pasang kembali guard pada `POST /api/internal/reset-all-tenants-to-free` sebelum non-dev.**
- ☐ FE: hapus pengiriman `requestedStorageGb` di checkout.
- ☐ FE: sembunyikan opsi self-serve untuk tier `CUSTOM` .
- ☐ FF: tangani error kuota workspace `400` .